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Theoretical Split-Window Algorithms for Determining the Actual Surface Temperature.

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Summary. – Two split-window strategies for obtaining the actual surface temperature (AST) from an infra-red sensor system are investigated by means of simulations of radiometer signals for a wide range of different geophysical situations. The differences between strategies are closely related to the strong or weak presence of atmospheric effects in the split-window coefficient. It has been demonstrated that the strong presence makes it necessary to know the total atmospheric water vapour as well as the effective emissivities in the two thermal channels for applying the strong split-window method, whereas for the weak presence one only needs to know the emissivities. Simplified algorithms have been obtained from these methods when some of the input data are unknown. Likewise, included for all these algorithms is a rigorous evaluation of their accuracies that takes into account the uncertainties in the emissivity and the noise-equivalent temperature. In this way several algorithms are provided for determining the AST, leaving to the choice of the reader the algorithm that best responds to the required accuracy and the available input data.

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1. – Introduction.

Satellite remote sensing of surface temperature is of considerable importance for many environmental studies. However, the determination of this parameter from satellite radiance constitutes a difficult task, as it is necessary to correct the remote measurement for absorption and emission of atmospheric water vapour as well as for the effect of surface emissivity, the latter being generally different from unity and spectrally variable.

At the present time, the split-window equation is the most popular method to eliminate atmospheric water vapour effects. Thus, using this method, numerous studies have been done to determine sea surface temperature [1-6] and land surface temperature [7-10]. It is based on two radiometric measurements, carried out in the

atmospheric window situated between 10.5 and 12.5 μm, for which atmospheric-absorption properties are different; it can be written in the linear case as

$$T_0 = T_1 + A(T_1 - T_2) + B\,, \tag{1}$$

where T_0 is the actual surface temperature (AST), T_1 is the radiometric temperature for the more transparent channel, T_2 is the radiometric temperature in the second channel, A is a coefficient that depends on the water vapour transmittances in both channels, and B is a constant that takes into account the surface emissivity effect and the CO_2 and aerosol attenuation.

Unfortunately, most of the above-mentioned works present different values for the split-window coefficients A and B, and some do not inform the reader of their range of validity, whereupon it is difficult to choose a pair of these coefficients for each pratical situation.

The aim of this paper is therefore to give a whole set of practical split-window algorithms that permit one to obtain the actual surface temperature (AST) from two brightness temperatures measured from space in two channels of interest, in particular in the NOAA-11 AVHRR channels 4 and 5, as a function of the required accuracy and the available input data (total atmospheric water vapour and the channel emissivities).

This paper has been divided into two parts. In the first part, we present a split-window algorithm from a theoretical viewpoint, called the SSW (Strong Split-Window) algorithm, that shows a clear coexistence of the atmospheric and emissivity effects in its coefficients. We also present another split-window formulation, namely the WSW (Weak Split-Window) algorithm, that permits a more separate analysis of the two effects. In the second part, simulations of NOAA-11 AVHRR channels 4 and 5 using the computer code LOWTRAN-7 permit checking out the accuracy of the aforementioned algorithms as well as other simpler algorithms that we have derived from these. Finally, this part also gives a discussion of the results in terms of a sensitivity analysis.

2. – Theory.

From the radiative transfer equation for a cloud-free atmosphere under local thermodynamic equilibrium, the radiance measured in channel i of a satellite, at zenith angle θ, can be written as

$$B_i(T_i) = \varepsilon_i B_i(T_0)\tau_i(\theta) + R_i\uparrow(\theta) + \frac{1-\varepsilon_i}{\pi} R_i\downarrow_{\text{hem}}\tau_i(\theta)\,, \tag{2}$$

where $B_i(T_i)$ is the weighted Planck's function for channel i, ε_i is the effective surface emissivity in channel i, $\tau_i(\theta)$ is the channel i transmittance of the atmosphere at the zenith angle θ, $R_i\uparrow(\theta)$ is the upward radiance in the θ direction, and finally $R_i\downarrow_{\text{hem}}$ is the downward hemispherical radiance emitted by the atmosphere.

The first term on the right-hand side of eq. (2) represents the received radiance resulting from surface emission. The second term is the radiance emitted by the atmosphere towards the sensor, and the last term is the downward atmospheric radiance that is reflected by the ground surface towards the sensor.

Starting from eq. (2), it is possible to develop a single-channel equation that permits

relating, for an atmosphere with water vapour as the only absorbing gas, the actual surface temperature, T_0, with the temperature measured by thermal infra-red channels, T_i. This model, obtained by linearization of both Planck's function and the atmospheric trasmittance, is based on [11]:

$$T_0 - T_i = \frac{1-\varepsilon_i}{\varepsilon_i} L_i + \frac{k_i W(T_i - T_a)}{\varepsilon_i \cos\theta (1 - k_i W/\cos\theta)} - 2\frac{1-\varepsilon_i}{\varepsilon_i} k_i W(T_a + L_i - T_i), \tag{3}$$

where L_i is a parameter with dimension of temperature defined as T_i/n_i, n_i being a constant for a given spectral channel, and with T_a, W and k_i being, respectively, the average atmospheric temperature, the total water vapour content of the atmosphere, and the effective absorption coefficient of the whole atmosphere in channel i. Values of these parameters are given in table I for 4 different atmospheric models.

TABLE I. – *Values of the total amount of water vapour, W (g cm^{-2}), the effective air temperature, T_a(K), the atmospheric absortion coefficients for channels* 4 *and* 5, k_4 *and* k_5 (cm^2/g), *the coefficients* a_0, a_1 *and* a_2 *of eq.* (6) *and the coefficients* b_4 *and* b_5 *of eq.* (7) *for the following atmospheric models*: (T) *tropical*, (MLS) *midlatitude summer*, (US), 1976 *standard* USA, *and* (MLW) *midlatitude winter.*

Model	W	T_a	k_4	k_5	a_0	a_1	a_2	b_4	b_5
T	3.32	290.6	0.122	0.171	2.49	4.33	–5.16	0.08	0.03
MLS	2.36	286.4	0.113	0.166	2.13	3.18	–7.49	0.22	0.06
US	1.13	277.1	0.092	0.146	1.71	1.30	–9.10	0.41	0.22
MLW	0.69	265.8	0.088	0.144	1.57	0.76	–9.08	0.45	0.26

Although eq. (3) itself constitutes a single-channel correction method, the presence of T_a creates an inconvenience that makes atmospheric radiosoundings necessary in the study area, which are rather the exception. Nevertheless, as can be seen further on, this problem can be avoided if two radiative temperatures are involved in the process of estimating the AST.

2·1. *Strong coexistence of the atmospheric and emissivity effects. The* SSW *algorithm.* – Using two channels (in particular the NOAA-11 AVHRR channels 4 and 5), the influence of T_a can be eliminated and eq. (1) is obtained. Thus, the atmospheric and emissivity correction can be given from the split-window coefficients A and B as [11]

$$A = \frac{\alpha_5\beta_4 + \beta_4\beta_5 W}{\gamma}, \tag{4}$$

$$B = \frac{1-\varepsilon_4}{\varepsilon_4}\frac{\alpha_4\beta_5}{\gamma}[1 - 2k_4 W]L_4 - \frac{1-\varepsilon_5}{\varepsilon_5}\frac{\alpha_5\beta_4}{\gamma}[1 - 2k_5 W]L_5, \tag{5}$$

with $\alpha_i = \varepsilon_i(\cos\theta - k_i W)$, $\beta_i = k_i\{1 + 2\alpha_i[(1-\varepsilon_i)/\varepsilon_i]\}$, and $\gamma = \alpha_4\beta_5 - \alpha_5\beta_4$.

Equations (4) and (5) are clearly influenced by the atmosphere (through W, k_i) and by the emissivity (through ε_i). Nevertheless, these influences are different in the two equations: thus the coefficient B is very sensitive to changes in the aforementioned parameters, whereas the coefficient A is only affected by the atmosphere, being

practically independent of changes in emissivity. As a result it is not possible from eqs. (4) and (5) to have a universal split window with sufficient accuracy, being it necessary to obtain a couple of these coefficients for each atmospheric situation and effective surface emissivity. To take this into account we have linearized both coefficients, for each atmospheric model, in terms of the channel 4 emissivity, ε_4, and the channel emissivity difference, $\Delta\varepsilon = \varepsilon_4 - \varepsilon_5$, as

$$A = a_0 + a_1(1-\varepsilon_4) + a_2\,\Delta\varepsilon\,, \tag{6}$$

$$B = T_4\left[\frac{1-\varepsilon_4}{\varepsilon_4}b_4 - \frac{1-\varepsilon_5}{\varepsilon_5}b_5\right], \tag{7}$$

where a_0, a_1, a_2, b_4 and b_5 are five parameters that are independent of the observation angle and express the influence of the existing atmosphere between the ground and the sensor (through the total water vapour content W). Thus, from table I, in which the values of the different atmospheric parameters are given, it is clear that a_0, a_1 and a_2 increase linearly with W, while b_4 and b_5 decrease linearly with W. At this point, it should be noted that we have eliminated in eq. (7) the terms on $\Delta\varepsilon$ included in equation (32) of Sobrino *et al.* [11], because this only contributes an error of less than 0.1 °C to the surface temperature.

These results show that for determining the AST from this SSW algorithm it is necessary to know the amount of total water vapour, W (so it is possible to obtain the appropriate parameters by interpolation in table I), and the effective emissivities in the thermal channels, ε_4 and ε_5. More discussion of the performance of this algorithm and others derived from it are given in sect. **3**.

2*2. *Towards a separated evaluation of the atmospheric and emissivity effects. The* WSW *algorithm.* – It has been shown that eqs. (6) and (7) can be used for obtaining the AST; however, they require knowledge of the total atmospheric water vapour, which can constitute a problem for many practical purposes. Writing eq. (3) for the AVHRR channels 4 and 5, we develop a new procedure that, combining statistical and physical approaches, permits solving this problem using only satellite data. It is based on the following relationship:

$$(T_0 - T_4) = P + Q\,, \tag{8}$$

where P and Q are two parameters that have been chosen to include, respectively, the greater part of the atmospheric influence and the greater part of the emissivity effects, and which are given by

$$P = (T_4 - T_5) + 2(\delta_4 - \sigma_4)(T_4 - T_a) - (\delta_5 - \sigma_5)(T_5 - T_a) + \varrho\,, \tag{9}$$

$$Q = 2\frac{1-\varepsilon_4}{\varepsilon_4}L_4 - \frac{1-\varepsilon_5}{\varepsilon_5}L_5\,, \tag{10}$$

with $\delta_i = \{k_i W/[\varepsilon_i(\cos\theta - k_i W)]\}$, $\sigma_i = 2k_i W[(1-\varepsilon_i)/\varepsilon_i]$ and $\varrho = \sigma_5 L_5 - 2\sigma_4 L_4$.

To assess these parameters, simulations of NOAA-11 AVHRR channels 4 and 5 measurements have been carried out using the LOWTRAN-7 atmospheric transmittance-radiance computer code [12] with the appropriate filter functions. The simulations were made for a group of model atmospheres representative of the variability of the atmospheric temperature and water vapour structure: the tropical

(T_0 = 299.7 K is the first boundary layer temperature of the atmosphere), the midlatitude summer (T_0 = 294.2 K), the midlatitude winter (T_0 = 272.2 K), and the 1976 US standard T_0 = 288.2 K) model atmospheres. Three observation angles (0 °, 30 ° and 50 °), three surface temperatures (T_0, T_0+5 and T_0-5) and 39 different combinations of channel 4 and 5 emissivities (ε_4 and ε_5 ranging from 0.96 to 1 and five values of the emissivity differences, $\Delta\varepsilon$, ranging from 0.01 to –0.01), were also considered for each atmosphere. This group of surface emissivities permits evaluating the influence of the atmospheric and emissivity effects for the vast majority of natural surfaces observed by remote sensing (bare soil, vegetation, water, or any combination of them, in a pixel). In this way, the radiometric temperatures were obtained using eq. (2) and inverting Planck's function for a wide range of different geophysical situations (1404 for each channel).

At this point, the following four considerations with respect to eq. (9) are necessary to understand the method and thus to give the parameter P as solely a function of the satellite measurements:

1) For our purposes, eq. (9) is not practical in this actual form because it includes, in addition to the channel emissivities, several atmospheric parameters (T_a, W, and k_i) for the knowledge of which it is necessary to have external measurements in addition to those given by the satellite (*viz.* atmospheric radiosoundings in the study area).

2) Using the atmospheric parameters included in table I, we have found and checked numerically, from simulations of radiometer signals, and especially for dry atmospheres, a good quadratic relationship between the right-hand side of eq. (9) and the difference of brightness temperature in the AVHRR channels 4 and 5, T_4-T_5. The problem is that this relationship is poor for wet atmospheres (midlatitude summer and tropical) with high surface temperatures (> 294.2 K and 299.7 K, respectively) and for large observation angles (> 40 ° and 30 °, respectively).

3) For ameliorating the results in these extreme situations it is necessary to include a second-order term in the linearizations of both Planck's function and atmospheric transmittance. This, however, entails an inconvenience because it complicates eq. (3) excessively.

4) To solve this drawback we propose a statistical procedure that gives excellent correlations with T_4-T_5 for all of the simulated cases. This procedure consists in evaluating the parameter P from $P=[(T_0-T_4)-Q]$.

Therefore, with these considerations and from the definition of Q, eq. (10), it is easy, using the simulated temperatures and the values of the constants for the AVHRR channels 4 and 5, n_4 = 4.673 and n_5 = 4.260 [11], to obtain the coefficient Q and so to plot T_0-T_4-Q *versus* T_4-T_5. In this way the parameter P was found, according to the following quadratic relationship:

$$P=[0.96+0.48(T_4-T_5)](T_4-T_5)\,, \tag{11}$$

which yields a correlation coefficient and a standard error of estimate of 0.975 and 0.40, respectively, and where the role of the split-window coefficient A (see eq. 1) is played by the quantities inside the square brackets. According to this, we have obtained a coefficient A that does not need the water vapour as an input parameter and depends

linearly on the temperature difference between the two channels, $T_4 - T_5$, which at the same time is a measure of the influence of the total water vapour content of the atmosphere, the observation angle, the surface temperature, and the emissivity.

Moreover, as is suggested by eq. (10), Q depends only on the channel emissivities ε_4 and ε_5, and not on the atmospheric state. In order to illustrate this dependence more clearly, it is possible to express Q from eq. (10) in terms of the following combination of ε_4 and $\Delta\varepsilon = (\varepsilon_4 - \varepsilon_5)$

$$Q = 58(1 - \varepsilon_4) - 70\,\Delta\varepsilon\,, \tag{12}$$

where 58 and 70 are two coefficients obtained from the multiple regression of the coefficient Q *versus* the channels 4 and 5 emissivities, with 0.998 and 0.06 K being, respectively, the correlation coefficient and standard error of estimate of Q. So, from the comparison between the AST used in the LOWTRAN-7 simulations and that given by eqs. (11) and (12), a standard error of 0.40 K is obtained.

In spite of these good results, we have made some modifications to eqs. (9) and (10) in order to give a better separation between the atmospheric and emissivity effects, and thus improve the accuracy of the SSW algorithm. Before describing these modifications, it is necessary to recall that the total separation of atmospheric and emissivity effects is not possible because radiometric temperatures are evidently disturbed by both effects. However, this is not a handicap for the improvement of this split-window algorithm. Thus, although the emissivity is present in every term of eq. (9), only the last term, ϱ, is strongly influenced by variations of this parameter. As an example of this, for a tropical model with $T_0 = 304.7$ K and $\theta = 0\,°$, and considering the 39 different combinations of channel 4 and 5 emissivities, the ϱ values yield a maximum difference of 2.7 K and a standard deviation of 0.72 K. Whereas for the other terms, that is $P - \varrho$, the emissivity influence only gives a maximum difference and a standard deviation of 0.55 K and 0.16 K, respectively. The most natural solution to this problem consists in introducing this last term in the coefficient Q and thus rewriting eqs. (9) and (10) in accordance with this modification. Neither solves the problem, however, because in this situation, although we have verified that a better correlation is obtained for the coefficient P (0.987), the coefficient Q (that in this case was obtained from the absorption coefficients given by table I) is more poorly correlated (0.922) with the emissivity of channels 4 and 5, which makes it difficult to evaluate the emissivity influence. These results are to be expected, however, because the term ϱ is a mixed term that takes into account the effects of both the atmosphere and the emissivity, therefore the previous modification only supposes to move the problem from the coefficient P to the coefficient Q. Then, for solving this difficulty, the best solution consists in separating both effects from the term ϱ.

To do this we propose to identify the emissivity effect with the term ϱ when obtained for the most transparent atmosphere. This procedure permits removing the majority of the emissivity influence ($\varrho - \Delta n$) on the coefficient Q, in which we have included the term ϱ for the midlatitude winter atmosphere, and therefore to keep the atmospheric influence, Δn, in the coefficient P. Evidently this new posing makes it necessary to modify eqs. (9) and (10). Thus, we rewrite these as

$$P = (T_4 - T_5) + 2(\delta_4 - \sigma_4)(T_4 - T_a) - (\delta_5 - \sigma_5)(T_5 - T_a) + \Delta n\,, \tag{13}$$

$$Q = 53(1-\varepsilon_4) - 53\,\Delta\varepsilon\,, \tag{14}$$

with

$$\Delta n = \left[2\frac{1-\varepsilon_4}{\varepsilon_4}L_4(1-2K_4W) - \frac{1-\varepsilon_5}{\varepsilon_5}L_5(1-2K_5W)\right] - [53(1-\varepsilon_4) - 53\,\Delta\varepsilon]\,. \tag{15}$$

Analogously to eq. (12), eq. (14) was obtained by multiple regression of the coefficient Q *versus* the channel 4 and 5 emissivities. We found a correlation coefficient of 0.999 and a standard error of estimate of 0.03 K. Likewise, here again, we have obtained P from the plot of $[(T_0 - T_4) - Q]$ *versus* $(T_4 - T_5)$ (see fig. 1). Then we propose from eqs. (14) to (16) a new split-window equation:

$$T_0 = T_4 + [1.060 + 0.46(T_4 - T_5)](T_4 - T_5) + 53(1-\varepsilon_4) - 53\,\Delta\varepsilon\,, \tag{16}$$

which gives the AST with a standard error of 0.33 K, which is lower than that given by the split-window algorithm obtained from eqs. (11) and (12).

Equation (16) constitutes the split-window model that we propose to obtain the AST. The importance of this algorithm lies in the fact that it is applicable at global scale if the surface emissivity is known, which constitutes a great advantage in comparison with the SSW algorithm. Here, the coefficient A (given by the quantities included by the square brackets, is strongly dependent on the atmospheric effects (through $T_4 - T_5$), and we need not know the total atmospheric water vapour; furthermore, the coefficient B (the last two terms on the right-hand side) only depends on the emissivity (through ε_4 and $\Delta\varepsilon$). From this coefficient, evaluation of the errors due to uncertainties in the knowledge of the surface emissivity is easy; thus, indeterminacy of 0.005 in ε_4 and ε_5 produces errors of 0.60 K in the AST.

Finally, we give in sect. **3** the accuracy of this method and of the other physical-statistical algorithms derived from it.

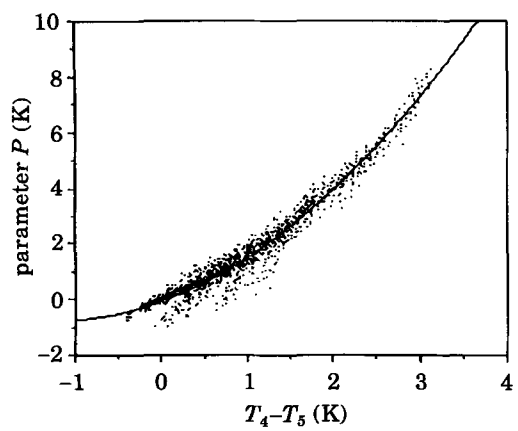

Fig. 1. – Plot of $P = [(T_0 - T_4) - Q]$ *vs.* $(T_4 - T_5)$ for the simulations considered (see eq. (16)). The solid line is the quadratic regression of the points. $r = 0.984$, SSE= 0.32 K, N=1404.

3. – Results.

The present section is divided into three parts. In the first part, we discuss some considerations with regard to the input parameters in the split-window algorithms. The second part presents a whole set of practical split-window algorithms derived from the procedures given in the previous section (the SSW and WSW algorithms) for use when some of the input data are unknown. Finally, part three includes a rigorous study and discussion of the accuracy of all these split-window algorithms.

3·1. *Input parameters in the split-window algorithms.*

3·1.1. Total atmospheric water vapour. – At this stage, it is clear that the main source of error for practical applications of eqs. (6) and (7) is ignorance of the total water vapour content of the atmosphere. In recent years, numerous methods have been developed to determine this parameter from satellite-based sounding radiometers such as the IRIS (Infra-red Interferometer Spectrometer) on board the NINBUS-4 satellite [13], the HIRS (High-resolution Infra-red Radiation Sounder) that is the infra-red part of TOVS (TIROS-N Operational Vertical Sounder) on board the NOAA satellite [14], the VISSR (Visible and Infra-red Spin Scan Radiometer) atmospheric sounder (VAS) on board the GOES satellite [15], etc. In practice, however, especially when the AVHRR data are the only ones available, it is more useful to obtain the water vapour from methods that use the linear relationship exhibited between the precipitable water vapour and the difference of radiometric temperatures in two infra-red channels. For this purpose, Dalu [16] gave a method that permits obtaining the water vapour with an error of 0.5 g/cm^2 using the AVHRR channels 4 and 5 temperatures. From radiative transfer simulations, Schluessel [17] estimated the water vapour with an error of 0.48 g/cm^2 at nadir view to 0.59 g/cm^2 at 50 scan angle for a single AVHRR pixel. Also Eck and Holben [18] estimated the precipitable water vapour from the $T_4 - T_5$ difference in the African Sahel and over areas of aerosol optical thicknesses less tham 0.4 and found an error of 0.55 g/cm^2.

In this way, practical methods that avoid the use of atmospheric radiosoundings can be applied for situations in which it is a sufficient approximate evaluation of the water vapour content, and therefore of the AST, for instance in the algorithms 2.1 and 2.2 of tables II and III.

3·1.2. Effective emissivities in AVHRR channels 4 and 5. – Except for the sea, which can be considered as a blackbody, the knowledge of the surface emissivity has a great importance for achieving an accurate evaluation of the AST. Thus, in our context, an indetermination of 0.005 in this parameter gives considerable errors in the AST (higher than 0.5 K) for the two split-window equations developed. Unfortunately, simultaneous access from satellite data to both T_0 and ε is not possible without some additional knowledge on ground surface radiative properties. To give this information numerous studies have been made [19-22, 9]. Although these results will not be related here, it is important to point out that all of them are in agreement with the absolute necessity of field emissivity measurements. On the other hand, although *in situ* measurements of ground emissivity have a good absolute accuracy, which is very difficult to obtain, other problems must be considered when these measurements are applied to satellite data, such as: 1) the bandwidth of the satellite channels, which makes a field measurement in the same wavelenghts necessary;

TABLE II. – *Input parameters and split-window coefficients A and B for the following split-window algorithms developed in the present paper*: 1 *obtained from eq.* (6) *and* (7), *and* 2.1, 2.2, 3, 4, 5, 6 *and* 7, *respectively, defined by eqs.* (17), (18), (19), (16), (21), (22) *and* (23). *The superscripts* (′) *given in the parameters of the coefficients A and B of the first algorithm indicate that they should be evaluated as a function of the total water vapour, that is by interpolation in table I. The superscript* (*) *of the total water vapour content included in algorithms* 2.1 *and* 2.2 *indicates that these algorithms only need an approximate evaluation of this parameter* (*see subsect.* **3**·1).

Model	Input data	Split-window coefficients	
		A	B
1	W, ε_4, ε_5	$a_0' + a_1'(1-\varepsilon_4) + a_2'\Delta\varepsilon$	$T_4\left[\frac{1-\varepsilon_4}{\varepsilon_4}b_4' - \frac{1-\varepsilon_5}{\varepsilon_5}b'5\right]$
2.1	W^*, ε_4, ε_5	$1.6+(1-\varepsilon_4)-9\Delta\varepsilon$	$55(1-\varepsilon_4)-70\Delta\varepsilon$
2.2	W^*, ε_4, ε_5	$2.3+3.8(1-\varepsilon_4)-6.3\Delta\varepsilon$	$33(1-\varepsilon_4)-14\Delta\varepsilon$
3	ε_4, ε_5	$2.0+2.4(1-\varepsilon_4)-7.7\Delta\varepsilon$	$44(1-\varepsilon_4)-42\Delta\varepsilon$
4	ε_4, ε_5	$1.06+0.46(T_4-T_5)$	$53(1-\varepsilon_4)-53\Delta\varepsilon$
5	ε_4	$0.61+0.59(T_4-T_5)$	$64(1-\varepsilon_4)$
6	$\varepsilon(10.5 \div 12.5)$ μm	$0.53+0.62(T_4-T_5)$	$64(1-\varepsilon_4)$
7	none	$0.36+0.65(T_4-T_5)$	1.40

2) the pixel dimension, composed often of a mixture of several surfaces (ground, vegetation, road, forest, etc.), which has made necessary the introduction of the effective emissivity [23-26]; and 3) the temporal variability of the target, which evidently alters the effective emissivity. However, in spite of these problems, Sobrino and Caselles [10] and Caselles and Sobrino [27] showed, for a vegetative surface that they know very well (an orange grove), how a meticulous analysis of the target observed by remote sensing complemented with the field measurements makes it possible to give an effective surface emissivity with a good accuracy (below 0.01). From this last result, it is clear that for obtaining good emissivities that can be used in practical situations, much more experimental work over a variety of surfaces is still necessary.

3·2. *Simplified split-window algorithms.*

3·2.1. From the SSW algorithm. – First we present the physical split-window algorithms, derived from eqs. (6) and (7), that we have called the SSW algorithm.

A) The winter algorithm.

This algorithm has been obtained from the average of the coefficients for the midlatitude winter and US standard model atmospheres shown in table I, and it is given by

$$T_0 = T_4 + [1.64 + (1-\varepsilon_4) - 9\,\Delta\varepsilon](T_4 - T_5) + 55(1-\varepsilon_4) - 70\,\Delta\varepsilon\,, \tag{17}$$

where the last two terms have been obtained from a multiple regression of the values of the coefficient B (see eq. 7), computed from 702 simulated channel temperatures for the aforementioned atmospheric models, *versus* the AVHRR channels 4 and 5 emissivities.

For this regression, a correlation coefficient of 0.998 and a standard deviation of 0.06 K were obtained.

B) The summer algorithm.

As in the previous algorithm, an average of the coefficients given in table I for the midlatitude summer and tropical model atmospheres permits writing

$$T_0 = T_4 + [2.31 + 3.8(1-\varepsilon_4) - 6.3\,\Delta\varepsilon](T_4 - T_5) + 33(1-\varepsilon_4) - 14\,\Delta\varepsilon\,, \tag{18}$$

with a correlation coefficient of 0.999 and a standard deviation of 0.02 K for the two last terms.

C) Worldwide algorithm.

In this case the average has been made for the 4 atmospheric models considered (see table I); then we yield

$$T_0 = T_4 + [1.98 + 2.4(1-\varepsilon_4) - 7.7\,\Delta\varepsilon](T_4 - T_5) + 44(1-\varepsilon_4) - 42\,\Delta\varepsilon\,, \tag{19}$$

0.998 and 0.05 being, respectively, the correlation coefficient and the standard deviation for the multiple regression. The importance of this algorithm lies in the fact that this is the only algorithm derived from the SSW that does not need the water vapour as an input parameter.

3·2.2. From the WSW algorithm. – Now, we present the split-window algorithms derived, like the WSW algorithm, from a combination of physical and statistical approaches.

A) Single emissivity measurement.

a) The emissivity of the AVHRR *channel* 4.

This interesting algorithm permits determining the AST when the channel 4 emissivity is the only input parameter known; it is based on the next relationship:

$$Q = L_4 \frac{1-\varepsilon_4}{\varepsilon_4}\,, \tag{20}$$

which permits, by simple regression with the channel 4 emissivity, obtaining Q as $64(1-\varepsilon_4)$ (see fig. 2), with a correlation coefficient of 0.998 and a standard error of estimate of 0.05 K. Then, the same procedure of the subsect. (**2**·1), that is to plot $(T_0 - T_4 - Q)$ *versus* $T_4 - T_5$, yields

$$T_0 = T_4 + [0.610 + 0.59(T_4 - T_5)](T_4 - T_5) + 64(1-\varepsilon_4) \tag{21}$$

for the data used, with a correlation coefficient of 0.966 and a standard error of estimate of 0.44 K for the quadratic regression (see fig. 3).

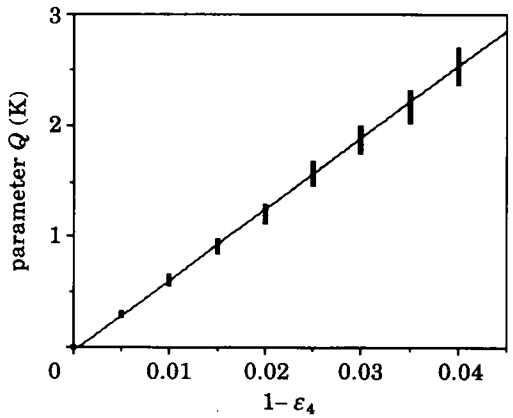

Fig. 2.

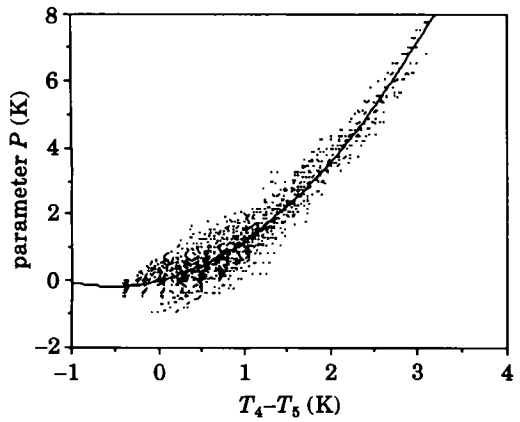

Fig. 3.

Fig. 2. – Plot of $[L_4(1-\varepsilon_4)/\varepsilon_4]$ *vs.* ε_4 (see eq. (21)). The solid line is the linear regression of the points. $r = 0.998$, SSE = 0.05 K, $N = 1404$.

Fig. 3. – The same as fig. 1 for eq. (21). $r = 0.966$, SSE = 0.44 K, $N = 1404$.

b) *The mean emissivity.*

Analogously to the above algorithm, this algorithm permits determining the AST when the mean effective emissivity, $\varepsilon = [(\varepsilon_4 + \varepsilon_5)/2]$, is the single input parameter known. It is based on the following relationship:

$$T_0 = T_4 + [0.530 + 0.62(T_4 - T_5)](T_4 - T_5) + 64(1-\varepsilon)\,, \tag{22}$$

which presents for Q the same correlation coefficient and standard error of estimate as that given by eq. (20), and for the quadratic regression, 0.942 and 0.58 K, respectively.

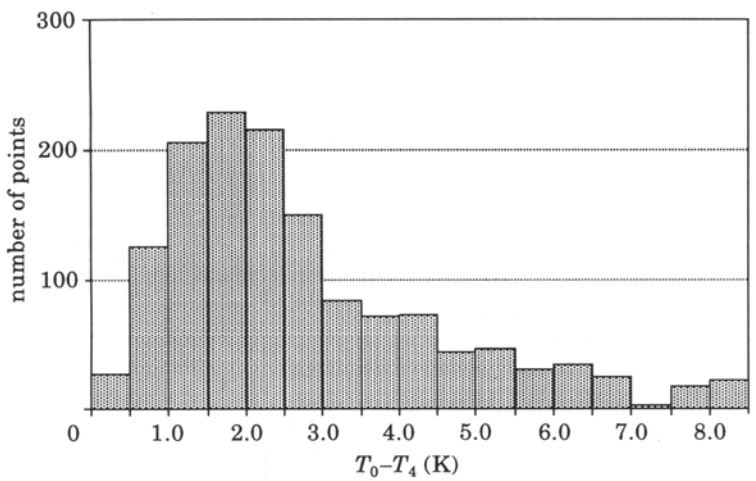

Fig. 4. – Frequency histogram of the simulated values $(T_0 - T_4)$ (K) that have been used for validating the different split-window algorithms.

The importance of eq. (22) lies in the fact that it permits obtaining the AST from *in situ* measurements of emissivity, which are generally made in the 10.5 to 12.5 μm wave band.

3·2.3. Without any input parameter. – This last algorithm, the only one obtained from the use of statistical approaches only, has been found by plotting $T_0 - T_4$ *versus* $T_4 - T_5$ for the cases considered, which permits writing

$$T_0 = T_4 + [0.360 + 0.65(T_4 - T_5)](T_4 - T_5) + 1.40\,, \tag{23}$$

0.904 and 0.75 K being, respectively, the correlation coefficient and the standard error of estimate for the quadratic regression. In spite of the elevated error with this algorithm, it can be very useful in many situations. Thus, for the 1404 simulated cases, eq. (23)

TABLE III. – *Standard error of estimate* (SSE) *in* K *of the split-window algorithms given in table II for each atmospheric model considered, for the whole of data (total), and excluding the tropical atmosphere* (no T.). *The square brackets include the minimum and maximum differences* (K) *between the* AST *and that given by the corresponding algorithm. The last column* (∂T_0) *indicates the percentage of values that present a difference less than* 0.5 K *for* T_{0L}–T_{0sw} *(see text). Finally, the latter file has been obtained if no correction is made to the remote-sensing measurement, that is,* $T_0 = T_4$.

	Atmospheric model						
Model	MLW	US	MLS	T	Total	No T.	∂T(%)
1	[–0.4, 0.0] SSE = 0.13	[–0.4, 0.3] SSE = 0.17	[–1.1, 0.3] SSE = 0.51	[–1.4, 0.6] SSE = 0.62	SSE = 0.41	SSE = 0.32	77
2.1	[–0.4, 0.1] SSE = 0.13	[–0.4, 0.4] SSE = 0.16	— —	— —	SSE = 0.15	—	100
2.2	— —	— —	[–1.4, 0.6] SSE = 0.44	[–1.6, 1.0] SSE = 0.62	SSE = 0.54	SSE = 0.44	62
3	[–0.8, 0.8] SSE = 0.39	[–0.8, 0.8] SSE = 0.33	[–0.9, 0.7] SSE = 0.34	[–1.9, 2.2] SSE = 0.78	SSE = 0.50	SSE = 0.36	74
4	[–0.5, 0.3] SSE = 0.15	[–0.6, 0.8] SSE = 0.25	[–1.1, 0.8] SSE = 0.35	[–1.7, 0.7] SSE = 0.47	SSE = 0.33	SSE = 0.26	89
5	[–0.9, 0.7] SSE = 0.39	[–1.0, 1.3] SSE = 0.45	[–1.1, 1.4] SSE = 0.50	[–1.4, 0.9] SSE = 0.45	SSE = 0.45	SSE = 0.45	71
6	[–1.1, 1.1] SSE = 0.54	[–1.2, 1.7] SSE = 0.64	[–1.2, 1.8] SSE = 0.62	[–1.3, 1.2] SSE = 0.53	SSE = 0.58	SSE = 0.59	58
7	[–1.7, 1.5] SSE = 0.74	[–1.8, 2.3] SSE = 0.84	[–1.5, 2.1] SSE = 0.77	[–1.9, 1.7] SSE = 0.64	SSE = 0.75	SSE = 0.79	61
$T_0 = T_4$	[0.1, 2.9] SSE = 1.53	[0.6, 4.1] SSE = 2.34	[0.6, 5.6] SSE = 3.13	[1.6, 8.4] SSE = 5.00	SSE = 3.26	SSE = 2.43	—

produces in T_0 a maximum error of only 2.3 K (see row 7 of table III), which, according to fig. 4, is better than the errors obtained if no correction of the radiometric temperatures is made (see fig. 4, which shows the frequency histogram of $T_0 - T_4$).

3·3. *Performance and discussion of the split-window algorithms.* – To study the performance of each of the split-window algorithms that appear in this paper, which are summarized in table II, we have constructed two tables and three figures.

In table III we give the values of the differences between the AST used in LOWTRAN-7 simulations, T_{0L}, and that obtained from the algorithms developed in this paper, T_{0sw}. It includes, in columns 2 to 5 and for the four atmospheric models considered, the interval of variation of the difference between the two temperatures as well as their standard errors. These last data are also included for all the simulated data (see column 6) and for the case in which the tropical atmosphere is not considered (see column 7). Finally, to give a practical estimate of the range of validity of each algorithm, table III also shows in its column 8 the percentage of values that have a difference below 0.5 K for $T_{0L} - T_{0sw}$.

The analysis of this table shows that the WSW algorithm, number 4, gives the best results for obtaining the AST in terms of the total data, with a standard error of 0.33 K. The SSW algorithm, number 1, gives suitable results for dry atmospheres but not for wet atmospheres because, as we have already pointed out, in this last case the linearizations made for both Planck's function and atmospheric transmittance are not very appropriate. A confirmation of this fact can be observed in the simplified algorithms of summer and winter, numbers 2.1 and 2.2. It is also noticeable that the three last algorithms, numbers 5 to 7, have errors greater that the SSW algorithm, from 0.45 K to 0.75 K, which is because they use fewer input parameters.

At this point we have made an exhaustive error analysis to determine the sensitivity of the algorithms proposed. To assess this, it is necessary to add to the error of the method (∂T_m), given in table III, the errors due to the uncertainties in the knowledge of the surface emissivity and radiometric temperature ($\partial T_{0,\varepsilon,T}$), which are not negligible (at least 0.12 K for the AVHRR radiometric temperatures and 0.005 for the surface emissivity). Accordingly, table IV shows the total error of T_0, ∂T_0, which was obtained as

$$\partial T_0 = [(\partial T_m)^2 + (\partial T_{0,\varepsilon,T})^2]^{1/2} , \tag{24}$$

where $\partial T_{0,\varepsilon,T}$ is found from equation (1) as

$$\partial T_{0,\varepsilon,T} = [(\partial T_4)^2 + (\partial X)^2 + (\partial B)^2]^{1/2} , \tag{25}$$

with $X = A(T_4 - T_5)$ and where ∂X is calculated for algorithms 1 to 3, in which $A = a_0 + a_1(1 - \varepsilon_4) + a_2(\varepsilon_4 - \varepsilon_5)$, as

$$\partial X^2 = [(a_2 - a_1)^2 + a_2^2]\partial\varepsilon^2(T_4 - T_5)^2 + [A\partial(T_4 - T_5)^2] \tag{26}$$

and for algorithms 4 to 7, in which $A = c_0 + c_1(T_4 - T_5)$, as

$$\partial X = [c_0 + 2c_1(T_4 - T_5)]\partial(T_4 - T_5). \tag{27}$$

Equation (25) has been evaluated for two differences in brightness temperatures, 1 K and 3 K, and considering two errors in the knowledge of the surface emissivity, $\partial\varepsilon_i = 0.005$ and $\partial\varepsilon_i = 0.01$.

TABLE IV. – *Total error in* K *of the split-window algorithms given in table II for all the simulated data and considering two different errors in emissivity* ($\partial\varepsilon$) *and two values of the difference* T_4–T_5. *Except for the first model, in which the total error is obtained using the standard deviation given in table III for the midlatitude winter and tropical atmospheres, the error has been obtained considering the standard deviation given in column* 6 *of table III.*

	Total error in T_0(K)			
	$T_4 - T_5 = 1$K		$T_4 - T_5 = 3$K	
Model	$\partial\varepsilon_i = 0.005$	$\partial\varepsilon_i = 0.01$	$\partial\varepsilon_i = 0.005$	$\partial\varepsilon_i = 0.01$
1	0.8 – 0.8	1.5 – 0.8	0.8 – 0.8	1.6 – 0.8
2.1	0.8	1.5	0.8	1.5
2.2	0.7	0.9	0.8	0.9
3	0.8	1.2	0.8	1.2
4	0.7	1.3	1.0	1.5
5	0.7	0.9	1.0	1.1
6	0.8	1.0	1.1	1.2
7	0.9	0.8	1.1	1.1

A detailed analysis of table IV shows that the major emissivity influence in the total error occurs for the algorithms that include two-channel emissivity, that is, the SSW algorithms, especially when they are applied to dry atmospheres, and the WSW algorithm (which shows variations in the total error of 0.7 K and 0.5 K, respectively, for each variation of 0.01 in the values of ε_4 and ε_5). Against wet atmospheres (in which the effect of the emissivity is greatly compensated by the surface reflection of the atmospheric radiation), and for algorithms 5 to 7, in which a single (or no) emissivity measurement is included, these variations are practically (or totally) negligible. Likewise, table IV also shows that the major influence of $T_4 - T_5$ occurs for the physical-statistical algorithms, numbers 4 to 7, which showed an increment of the total error of 0.2 K for a variation of 2 K in the values of $T_4 - T_5$, while for the others this influence is negligible. It is also noticeable from this table that the introduction of the errors due to the emissivity and temperature totally changes the results given in table III especially for indeterminations of 0.005 in emissivity, situation in which a clear tendency towards the homogeneity of the errors is shown. Evidently these results can be surprising especially with respect to algorithms 5 to 7, and, therefore, lead to the following question: How is it possible that with fewer input parameters, ε_4 or ε, or even more using the only radiometric temperatures, the total error obtained can be the same or smaller (see column 4 of table IV) than that given when two emissivity measurements and the water vapour, if it is necessary, are considered? It must be noted that, in terms of the error theory, the answer to this question is clear: for these algorithms the impact of the knowledge of the emissivity in the determination of the AST (none for algorithm 7 and of 0.32 K and 0.64 K, respectively, for each variation of 0.005 and 0.01 in the values of ε_4 or ε for algorithms 5 and 6) is smaller than that associated with the WSW algorithm (0.70 K and 1.30 K, respectively for each variation

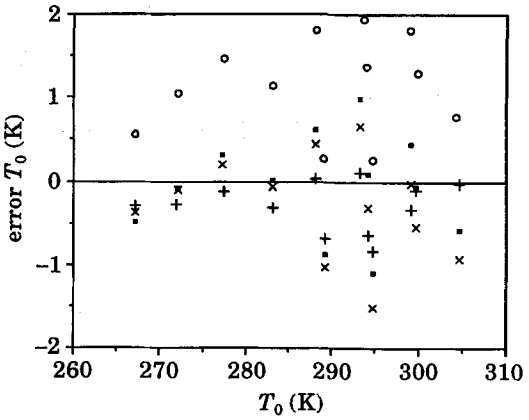

Fig. 5.

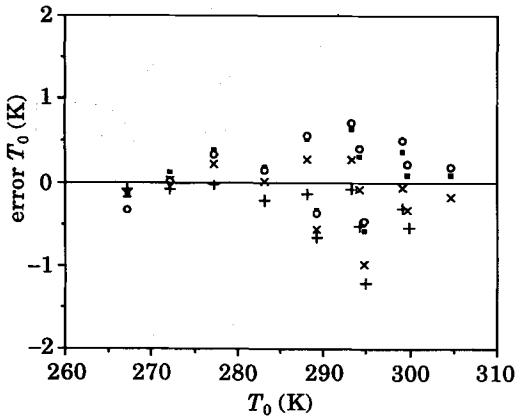

Fig. 6.

Fig. 5. – Differences (K) for a sandy surface ($\varepsilon_4 = 0.96$ and $\varepsilon_5 = 0.97$) between T_0 and that derived using algorithms 1, 4, 5 and 7 of table II over the corresponding true surface temperatures for an observation angle of 50°. The solid line shows the case in which no error is obtained. + 1, × 4, ■ 5, ○ 7.

Fig. 6. – Same as fig. 5 for a vegetated homogeneous surface (crop) ($\varepsilon_4 = 0.980$ and $\varepsilon_5 = 0.985$). Symbols as in fig. 5.

of 0.005 and 0.01 in the values of ε_4 and ε_5) and therefore this fact compensates for the errors shown in table III.

In spite of this justification, more considerations are necessary to understand the good performance of the last algorithm, number 7. Thus, it will be necessary to remember that this algorithm, obtained as a function of the simulated radiometric temperatures only, is strongly dependent on the characteristics of the simulated data. In this direction, although the algorithm shows a good performance for the whole of the 39 combinations of simulated emissivities it could happen that its performance be not very appropriate for one of the surfaces observed by remote sensing (see table III

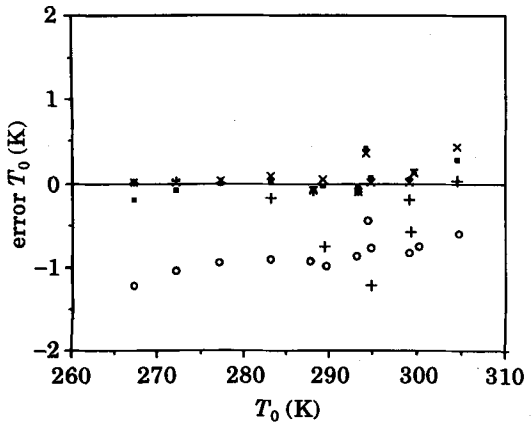

Fig. 7. – Same as fig. 5 for the sea surface ($\varepsilon_4 = 0.995$ and $\varepsilon_5 = 0.990$). Symbols as in fig. 5.

in which differences until 2.3 K in the AST are obtained for this algorithm). This problem, that also extends to all the algorithms based on statistical approaches, leads to the next reflection: «in environmental models, not only the root mean-square deviation between actual and predicted data should be considered, but also the maximun and minimum deviations between the two». Therefore, in this direction, two new basic questions must be answered: 1) Are all the combinations given for the emissivities of channels 4 and 5 real?; 2) How real are the *in situ* emissivity measurements when they are applied to the dimensions of a satellite pixel? The answers to these questions are very difficult because of the reasoning included in paragraph 3·1.2 and they will be only solved when the remote measurements permit acceding to the channel emissivities. Therefore, at this moment, the single possibility consists in assessing the peformance of every algorithm from *in situ* measurements of emissivity. To carry out this study we show in fig. 5 to 7, and for the four more characteristic algorithms (numbers 1, 4, 5 and 7 of table II) and three of the most usual surfaces observed by remote sensing, that is sand, vegetation and sea, the plot of the differences between true and retrieved values of the AST over the corresponding true surface temperatures (the twelve values of T_0 used in LOWTRAN simulations) for an observation angle of 50°. We have chosen these surfaces because they cover the interval of variation of the emissivity values (0.96 to 1) in the (10.5 ÷ 12.5) μm band, which permits checking the emissivity effect from three single figures. Thus, we observe the good performance of the algorithms considered, except algorithm 4 for a sandy surface and algorithm 7 for both the sea and the sand.

4. – Summary and conclusions.

Starting from the solution of the radiative transfer equation for an atmosphere with water vapour as the only absorbing gas, see eq. (2), we have developed two different procedures that permit correcting the temperatures given by thermal infra-red sensors. The first method developed, which has been called the Strong Split Window (SSW), due to the strong influence of atmospheric effects in its split-window coefficient B, is based on the work of Sobrino *et al.*, [11] and permits (from the knowledge of the total water vapour and the AVHRR channels 4 and 5 emissivities) obtaining the AST with a total error of 0.8 K for dry atmospheres and 1.5 K for wet atmospheres. The main problem of this method is that it needs accurate knowledge of the total water vapour, for which it is necessary to have data external to that provided by the satellites (for instance, radiosoundings that coincide with the satellite overpass). To solve this difficulty we have developed a second method that does not need water vapour as an input parameter. This method, called the Weak Split Window (WSW), due to the weak influence of the atmospheric effect in its split-window coefficient B, obtains the AST from the combinations of physical and statistical considerations with an error from 0.7 K (for $\partial\varepsilon = 0.005$ and $T_4 - T_5 = 1$ K) to 1.3 K (for $\partial\varepsilon = 0.01$ and $T_4 - T_5 = 3$ K). It is based on the linear relationship exhibited between the coefficient A and the difference $T_4 - T_5$ obtained for the simulations, which have been made using LOWTRAN-7 for a wide range of atmospheric temperature and humidity profiles representative of tropical, midlatitude summer and winter, and US standard atmospheres. This coefficient A can be applicable at global scale and only needs the remote measurements. On the other hand, this algorithm includes the major part of the emissivity influence of the split-window coefficient B, which has been obtained for the most transparent atmosphere. This

coefficient, which has been linearized in terms of the AVHRR channel 4 and 5 emissivities, permits an easy evaluation of the errors due to emissivity indeterminacy. Thus, we have found that variations of 0.005 in the values of ε_4 and ε_5 give the same variation (0.60 K) for B.

In addition to these algorithms, we have derived other much simpler algorithms that can be used with less input parameters. Thus, more precisely, we have shown that:

i) Estimations of total water vapour content from AVHRR data permit deriving from the SSW algorithm seasonal split-window algorithms with similar errors to the SSW algorithm (from 0.8 K to 1.5 K for dry atmospheres, and from 0.7 K to 0.9 K for wet atmospheres).

ii) Ignoring the total water vapour content, it is possible to develop from the SSW algorithm a worldwide split window that depends only on the channel emissivities, and which gives the AST with an error from 0.8 K (for $\partial\varepsilon = 0.005$) to 1.2 K (for $\partial\varepsilon = 0.01$).

iii) Using the same statistical procedure that was used to obtain the WSW algorithm, we have obtained a split-window method that needs to know only the channel 4 emissivity (or the mean emissivity), and which gives the AST with an error from 0.8 K (for $\partial\varepsilon = 0.005$ and $T_4 - T_5 = 1$ K) to 1.1 K (for $\partial\varepsilon = 0.01$ and $T_4 - T_5 = 3$ K).

iv) In extreme cases, when the input data are not known, a general expression that uses only the remote sensing temperatures gives the AST with an error from 0.9 K (for $T_4 - T_5 = 1$K) to 1.1 K (for $T_4 - T_5 = 3$ K).

Our results show the interest of the combination of physical and statistical approaches in order to exploit fully all the information contained in the measurements. In this way we have provided several tools for determining the AST, the algorithm that best responds to the required accuracy and the available input data remaining to the choice of the interested reader. Therefore, the present work constitutes an important step towards obtaining the AST from remotely sensed thermal infra-red measurements.

* * *

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